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Let $X \sim N(\mu, \sigma^2) \Rightarrow$ PDF, $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-1}{2\sigma^2}(x-\mu)^2}$, $-\infty < x < \infty$, $-\infty < \mu < \infty$, $0 < \sigma < \infty$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1, \quad E(X) = \mu, \quad V(x) = \sigma^2$$

CDF : $F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt = \Phi\left(\frac{x-\mu}{\sigma}\right)$, $\Phi(z) = \int_{-\infty}^{z} \phi(u)\,du$

$= \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\,du$

CDF of Std. normal function

Sampling from normal distribution

Let $X_1, X_2, \ldots, X_n$ be iid RVs from a $N(\mu, \sigma^2)$ distribution

↳ independent and identically distributed

$f(x_i) = N(\mu, \sigma^2)$

⊙ Population

↓

Total collection of elements of our interest under study ~~contributes~~ constitutes a population. A sample in any subset of the population under study.

Random Sample: Let $X_1, \ldots, X_n$ be n indep RVs taken ~~on~~ from the prob distribution with common PDF $f_X(x)$

~~to~~ then $X_1, X_2, \ldots, X_n$ is called a random sample of size n from the population whose prob distribⁿ is $f_X(x)$

iid ⟷ random sample

same

Then, joint PDF of $(X_1, \ldots, X_n)$

$$f_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} f_{X_i}(x_i)$$

$$\bar{X} = \frac{\Sigma X_i}{n}$$

Statistic : Any funcⁿ of $X_1, \ldots, X_n$; e.g.- sample mean, sample variance

$$S^2 = \underbrace{\frac{1}{n-1}}_{\text{will discuss later}} \sum_{i=1}^{\infty} (X_i - \bar{X})^2$$

Sampling Distribⁿ : Prob. distribⁿ of a statistics

Let $X_1, \ldots, X_n$ be a random sample from a normal $N(\mu, \sigma^2)$
what is the prob. distribⁿ of S.M, SV, etc.

Prob. Distribⁿ of $\bar{X}$ = ?

Let $Y = \bar{X}$, $E(Y) = E(\bar{X}) = E\left(\frac{X_1 + X_2 + \ldots + X_n}{n}\right)$

$$= \frac{1}{n}\left(\sum_{i=1}^{n} E(X_i)\right)$$

$$= \frac{1}{n}(\Sigma \mu) = \mu$$

$$V(Y) = V(\bar{X}) = V\left(\frac{1}{n} \sum_{i=1}^{n} X_i\right)$$

$(X_i \to \text{iid})$

$$= \frac{1}{n^2}(n\sigma^2) = \frac{\sigma}{n}$$

$$\bar{X} = Y \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$f_Y(y) = \frac{\sqrt{n}}{2\pi\sigma} e^{-\frac{n}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

Let $X \sim N(\mu, \sigma^2)$ and $Z = X-\mu/\sigma \sim N(0,1)$

if $Z^2 = \dfrac{(X-\mu)^2}{\sigma^2}$ then distribⁿ ?

if $X \sim G^{(r,\lambda)}$ (Gamma distribⁿ)

follows gamma distribⁿ $f_X(x) = \dfrac{\lambda^r x^{r-1} e^{-x\lambda}}{\Gamma(r)}$ ①, $x>0$, $\lambda>0$, $r>0$ $\rightarrow E(X) = r/\lambda$, $V(X) = r/\lambda^2$

also written as $\dfrac{x^{r-1} e^{-x/\lambda}}{\lambda^r \Gamma(r)}$ ② $\rightarrow E(X) = r\lambda$, $V(X) = r\lambda^2$

in ①, consider $r = n/2$, $\lambda = 1/2$
$(n \in \mathbb{N})$

$$f_X(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}$$ → here only one parameter n

PDF of a $\chi^2(n)$ R.V.
↳ chi squared

if $X \sim \chi^2(n) \Rightarrow f_X(x) = \dfrac{x^{\frac{n}{2}-1} e^{-\frac{x}{2}}}{2^{n/2} \Gamma(n/2)}$, $x>0$, $n \in \mathbb{N}$

$\epsilon \sim N(\mu, \sigma)$

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Statistics

If $X \sim \chi^2(n)$

$$f_X(x) = \frac{x^{n/2} e^{-x/2}}{2^{n/2} \Gamma(n/2)}, \quad x > 0$$

$E(X) = n$; $V(X) = 2n$

Let $X_1, \ldots, X_n$ be iid R.V.s sampled from a $\mathcal{N}(\mu, \sigma)$

Let $X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X-\mu}{\sigma} \sim N(0,1)$

$$Y = Z^2 = \left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$$

$$f_X(x) = \frac{e^{-x/2}}{\sqrt{2\pi x}}$$

C.D.F (Y) = $F_Y(y) = P(Y \le y) = P(Z^2 \le y)$

$$\text{CDF}(Y) = \begin{cases} P(|Z| \le \sqrt{y}), & y \ge 0 \\ 0, & y < 0 \end{cases}$$

$$F_Y(y) = \begin{cases} P(-\sqrt{y} \le Z \le \sqrt{y}), & y \ge 0 \\ 0, & y < 0 \end{cases}$$

$$= \begin{cases} F_Z(\sqrt{y}) - F_Z(-\sqrt{y}), & y > 0 \\ 0, & y < 0 \end{cases}$$

$$f_Y(y) = \frac{d\, F_Y(y)}{dy} = \frac{f_Z(\sqrt{y}) + f_Z(-\sqrt{y})}{2\sqrt{y}}, \quad y > 0$$

Given, $Z \sim N(0,1) \Rightarrow f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{\sqrt{y}}$

- If $Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2(1)$
  ↳ $X \sim N(\mu, \sigma^2)$ then $\left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi^2(1)$

- Consider two indep. r.v. X, Y s.t. $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2)$
  then $aX \pm bY \sim N(a\mu_1 \pm b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$

- Let $X \sim \chi^2(n_1)$, $Y \sim \chi^2(n_2)$, X & $Y \to$ indep.
  then $X+Y \sim \chi^2(n_1+n_2)$

- Consider $W = \sum_{i=1}^{n} \left(\frac{X_i-\mu}{\sigma}\right)^2 = \sum_{i=1}^{n} \chi^2(1)$ $(X_i \to iid)$
  $\Rightarrow W \sim \chi^2(n)$

- Prob'ty distribn of $S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$ not all terms are indep.

  $\Rightarrow \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$; $\bar{x} = \frac{\Sigma x_i}{n}$

  $E\left(\frac{(n-1)S^2}{\sigma^2}\right) = n-1$

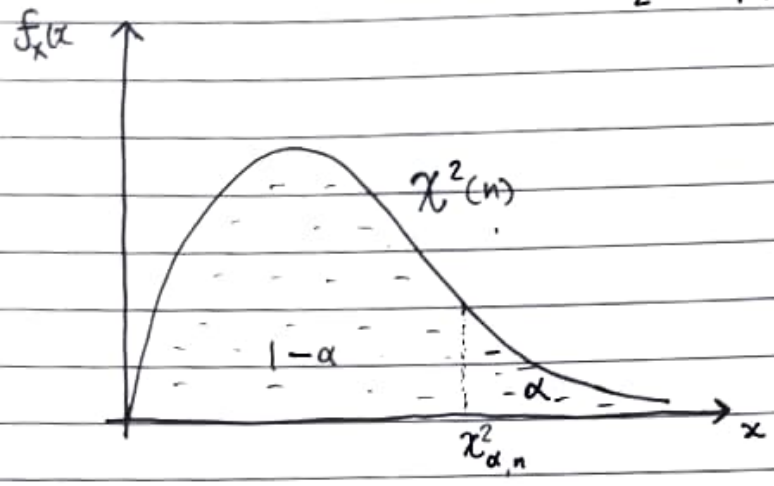
  $\Rightarrow E(S^2) = \sigma^2$ → that's why we defined $S = \frac{1}{n-1} \Sigma (x - x_i)^2$

  $V\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1)$

  $\Rightarrow V(S^2) = \frac{2\sigma^4}{(n-1)}$

  $\Rightarrow E(S^4) = \frac{2\sigma^4}{n-1} + \sigma^4 = \frac{n+1}{n-1}\sigma^4$

- If $X \sim \chi^2(n)$ then $f_X(x) = \dfrac{x^{n/2-1}\, e^{-x/2}}{2^{n/2}\, \Gamma(n/2)}$, $x > 0$, $n \in \mathbb{N}$

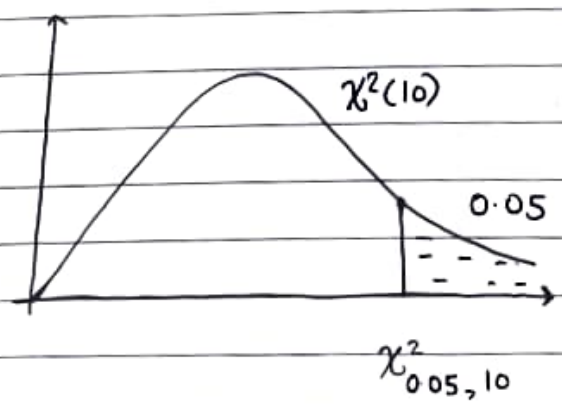


Upper α percentile : If $X \sim \chi^2(n)$. Then upper α-percentile of X is defined as $\chi^2_{\alpha,n}$

$$P(X > \chi^2_{\alpha,n}) = \alpha, \quad 0 < \alpha < 1$$
$$P(\chi^2(n) > \chi^2_{\alpha,n}) = \alpha$$

e.g.- Let $X \sim \chi^2(10)$. Find $\chi^2_{(0.05,10)}$

$$P(\chi^2(10) > \chi^2_{(0.05,10)}) = 0.05$$



From χ^2 table, we have $\chi^2_{0.05,10} = 18.306$

$\Rightarrow P(X > 18.306) = 0.05$ if $X \sim \chi^2(10)$

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Q.) $P(32.007 < X \leq \chi^2_{\alpha,23}) = 0.09$, $\chi^2_{\alpha,23} = ?$

$$P(X > 32.007) - P(X > \chi^2_{\alpha,13})$$
$$0.1 - P(X > \chi^2_{\alpha,23}) = 0.09$$
$$0.01 = P(X > \chi^2_{\alpha,13})$$
$$\alpha = 0.01$$
$$\Rightarrow \chi^2_{\alpha,23} = 41.638$$

Q.) $P(X > \chi^2_{\alpha,10}) - P(X > 23.209) = 0.015$
$$- 0.01 = 0.025$$
$$P(X > \chi^2_{\alpha,10}) = 0.025$$
$$\chi^2_{\alpha,10} = 20.483$$

- <u>T-distribn</u>

Let $X \sim N(0,1)$; $Y \sim \chi^2(n)$; X & $Y \rightarrow$ indep RV

$$T = \frac{X}{\sqrt{Y/n}}$$ then T follows a Student's t-distribn

$$f_T(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\,\Gamma\left(\frac{n}{2}\right)} \cdot \frac{1}{(1+t^2/n)^{n+1/2}}, \quad -\infty < t < \infty, \quad n \in \mathbb{N}$$

- Result :- $X_1, X_2, \ldots, X_n \sim N(\mu, \sigma^2)$. find prob distribn of $\sqrt{n}\left(\frac{\bar{X}-\mu}{S}\right)$

$$\bar{X} \sim N(\mu, \sigma^2/n) \Rightarrow \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma} \sim N(0,1)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

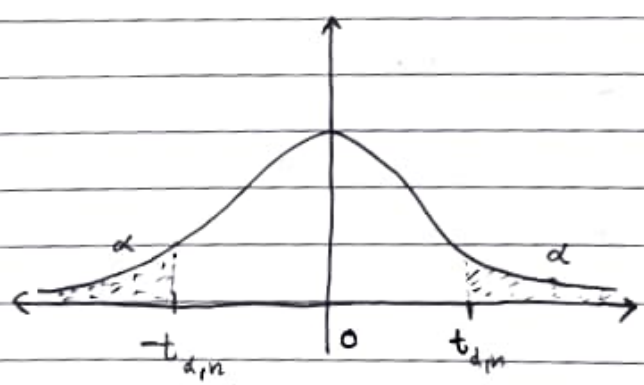
$$\frac{\frac{\sqrt{n}\,(\bar{X}-\mu)}{\sigma}}{\sqrt{\frac{1}{(n-1)}\left(\frac{(n-1)S^2}{\sigma^2}\right)}} \sim t(n-1)$$

$$\frac{\sqrt{n}\,(\bar{X}-\mu)}{S} \sim t(n-1)$$

$T \sim t(n) \rightarrow$ symmetric about $t=0$

Let $T \sim t(n)$, then α percentile of T is given by
$P(T > t_{\alpha,n}) = \alpha \quad (0<\alpha<1)$

$$t_{1-\alpha,n} = -t_{\alpha,n}$$



Q.) $P(-1.35 < T_{12} < 2.179)$

$P(T > -1.35) - P(2.179)$

$0.9 - 0.025 = 0.875$

Estimation Problems

- Suppose $X_1, X_2, \dots, X_n \rightarrow$ random sample taken from a prob distribn $f_X(x, \theta)$ where θ is an unknown parameter. Then the problem of estimation is to estimate θ on the basis of $X_1, X_2, \dots, X_n$

- Estimator $\rightarrow$ Any funcn of $X_1, X_2, \dots, X_n$ which is used to eliminate the unknown parameter θ is called an estimator of θ. $\rightarrow T(X_1, X_2, \dots, X_n) = T(\underline{X})$, $\underline{X} = (X_1, \dots, X_n)$

- Parameter Space $\rightarrow$ It is the set of all possible values of a parameter θ. It is often denoted by Θ (or Ω).

$$X \sim N(\mu, \sigma^2)$$

$\mu \rightarrow$ unknown, $\sigma \rightarrow$ known

$$\Theta : \{\mu : -\infty < \mu < \infty\}$$

if both unknown

$$\Theta : \{(\mu, \sigma^2) : -\infty < \mu < \infty, \sigma > 0\}$$

- Broadly estimation problems are classified as :-
  i) Point estimation
  ii) Interval estimation

  In a given problem, the interest could be to estimate θ or some funcn of θ, say $g(\theta)$.

- Point Estimation : Methods :-
  i) Unbiased estimation
  ii) Maxm likelihood estimation

<u>Def^n</u>: Let $X_1, \dots, X_n$ be a random sample (iid rvs) taken from a $f_X(x, \theta)$ dist^n where θ is an unknown paramer. An estimater $T(\underline{X})$ of θ is said to unbiased estimater of θ if $E(T(\underline{X})) = \theta$, $\forall\, \theta \in \Theta$.

For estimating $g(\theta)$, $E(T(\underline{X})) = g(\theta)\ \forall\, \theta \in \Theta$

e.g.- Let $X \sim Bin(n, p)$, n known, p unknown. Find an unbiased ~~parameter~~ estimator p.

Consider $T = \frac{X}{n} \Rightarrow E(T) = E(X/n) = \frac{1}{n} \cdot np = p$

$\Rightarrow T = \frac{X}{n}$ is an unbiased estimator of $p\ \forall\, p \in [0, 1]$

e.g.- Let $X_1, \dots, X_n$ be iid rv from a $Ber(p)$ distrib^n where p is an unknown parameter. Find unbiased estimator of P

Take $T = X_1$

$E(T) = E(X_1) = p$

Take $T_1 = \alpha X_1 + (1-\alpha) X_2$, $0 \le \alpha \le 1$

$E(T_1) = p$

$T^* = \frac{X_1 + \dots + X_n}{n}$

$E(T^*) = p$

$\boxed{\text{Let } X, Y \text{ be random indep rv, } V(X+Y) = V(X) + V(Y)}$

$V(T) = p(1-p)$

$V(T^*) = p(1-p)/n \Rightarrow V(T) \ge V(T^*)\ \forall\ n = 1, 2, \dots,\ 0 < p < 1$

$\Rightarrow T^*$ is a better unbiased estimator of p than T

⊙ If $X_1, X_2, \dots, X_n \overset{iid}{\sim} Ber(n, p)$ then $S = \Sigma X_i \sim Bin(n, p)$

—X—

e.g.- Let $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$

unbiased estimator of $\lambda = ?$

Some $T = X_1, \frac{X_1 + X_2 + X_3}{3}, \dots$

Among all these estimators, $T = \Sigma X_i / n$ has the min^m variance

$V(T(n)) = \lambda / n$

e.g.- not applicable for $\frac{1}{\pi} \cdot \frac{1}{1+x^2}$ coz $E(X)$ does not exis

⊙ Let $X_1, \dots, X_n$ be a random sample from a $f_X(x, \theta)$ dist^n.
If $E(X_i) = \theta$ exists for $i = 1, 2, \dots, n$ then sample mean $\bar{X}$ is always an unbiased estimator of population mean θ

e.g.- $X_1, \dots, X_n$ $X_i \sim N(\mu, \sigma)$ σ known, μ unknown

$E(\bar{X}) = \mu$, $V(\bar{X}) = \frac{\sigma^2}{n}$

($X_i \to$ iid from $f_X(x, \theta)$)

⊙ If $V(X_i) = \sigma^2$ for $i = 1, 2, \dots, n$ then sample variance is always an ~~ub~~ unbiased estimator of population variance (σ^2).

e.g.- $X_1, X_2, \dots, X_n \sim P(\lambda)$

$E(\bar{X}) = \lambda \to$ here also it is better than S^2

$E(S^2) = \lambda$

e.g.- $X \sim \text{Bin}(n, p)$, n known, p unknow

estimator of p^2

$$E(X(X-1)) = E(X^2) - E(X)$$
$$= V(X) + (E(X))^2 - E(X)$$
$$= np(1-p) + n^2p^2 - np$$

$$(n^2-n)p^2 = n(n-1)p^2$$

$$T = \frac{X(X-1)}{n(n-1)} \Rightarrow E(T) = p^2$$

Q) unbiased estimater of $V(X) = npq = np(1-p)$

$$T = \frac{X(n-X)}{(n-1)}$$

⊙ Unbiased estimator may not exist

↓

e.g.- $X \sim Bin(n,p)$ then estimator of $p^{n+1} = ?$

Assume $\exists\, T(X)$, $E(T(X)) = p^{n+1}$

$$\Rightarrow \sum_{x=0}^{n} T(x) \binom{n}{p} p^x (1-p)^{n-x} = p^{n+1}$$

$$\Rightarrow \underbrace{\sum_{x=0}^{n} T(x) \binom{n}{p} \left(\frac{p}{1-p}\right)^x (1-p)^n}_{} = p^{n+1}$$

LHS is a polynomial of deg $\leq n$. However $p^{n+1} \to (n+1)$th degree → $T(x)$ can't have p in it

↳ contradiction

↳ no unbiased estimator

⊙ Efficiency : Let T_1 and T_2 be two unbiased estimators of a parameter θ. Assume that $E(T_1^2) < \infty$, $E(T_2^2) < \infty$, we define efficiency of T_1 rel. to T_2

$$eff(T_1/T_2) = \frac{V(T_2)}{V(T_1)}$$

T_1 is more efficient than eff $(T_1/T_2) > 1$

e.g.- $X_1, \dots, X_n \sim N(\mu, 1)$, $\mu \to$ unknown parameter

$$T_1 = \bar{X} \Rightarrow E(T) = \mu \quad , \quad Var(\bar{X}) = 1/n$$

$$T_2 = \frac{X_1 + 2X_2 + \dots + nX_n}{n(n+1)/2} \quad , \quad E(T_2) = \mu$$

$$Var(T_2) = \frac{2(2n+1)}{3n(n+1)}$$

$$eff(T_1/T_2) = \frac{2}{3}\frac{(2n+1)}{(n+1)}$$

$$= \frac{2}{3}\left(1 + \frac{n}{n+1}\right)$$

$4n+2 \geq 3n+3$

$n \geq 1 \to$ always holds

Maximum Liklihood Estimation Method

- Suppose $X_1, \dots, X_n$ are iid RVs from a $f(x, \theta)$ distribⁿ where $\theta \to$ unknown.

Also let $\underline{x} = (x_1, x_2, \dots, x_n)$ be realized value of $\underline{X} = (X_1, \dots, X_n)$

Then likelihood fonᶜ of θ is defined as

$$L(\theta / \underline{x}) = L(\theta, \underline{x}) = \underbrace{\prod_{i=1}^{n} f_X(x_i, \theta)}_{\text{joint pdf of } X_1, \dots, X_n} \quad \text{— ①}$$

here it's only a funcⁿ of θ as $\underline{x}$ is known

- The value that maximizes $L(\theta / \underline{x})$ is called the most likelihood estimator of θ. often MLE of θ is denoted as $\hat{\theta}(\underline{x})$ (or just $\hat{\theta}$)

$$L(\hat{\theta}, \underline{x}) = \max_{\theta \in \Theta} L(\theta, \underline{x})$$

- This is the most widely used estimation method developed by R.A. Fischer (1922).

Q.) Let $X_1, \dots, X_n$ be iid Ber(θ) RV where p is an unknown parameter. Find MLE of θ.

$$P(X = x) = \theta^{x} (1-\theta)^{1-x} \quad ; \quad x = 0, 1 \; ; \; 0 < \theta < 1$$

$$\text{MLE} = \prod_{i=1}^{n} \theta^{x_i} (1-\theta)^{1-x_i}$$

$$= \theta^{\sum_{i=1}^{n} x_i} (1-\theta)^{n - \sum_{i=1}^{n} x_i}$$

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Now, log likelihood func^n of θ given $\underline{x}$ is

$$\ell_n(\theta) = \log L(\theta, \underline{x}) = \log\left(\theta^{\Sigma x_i}(1-\theta)^{n-\Sigma x_i}\right)$$

$$= (\Sigma x_i)\ln\theta + (n-\Sigma x_i)\ln(1-\theta)$$

maximizing $L(\theta)$ or $\log(L(\theta))$ is same thing

at maxim^m, $\dfrac{a}{\theta} - \dfrac{b}{1-\theta} = 0$ $\quad (a = \Sigma x_i,\ b = n - \Sigma x_i)$

$$\Rightarrow \frac{a}{a+b} = \theta \Rightarrow \hat{\theta} = \frac{\Sigma x_i}{n}$$

$$\left.\frac{\partial^2 \ell_n(\theta)}{\partial\theta^2}\right|_{\hat{\theta}} = \frac{-\Sigma x_i}{\theta^2} - \frac{(n-\Sigma x_i)}{(1-\theta)^2}$$

$$< 0 \quad \forall\ \theta \in (0,1)$$

We see that $\hat{\theta}$ is also an unbiased estimator of θ. In general MLEs are not unbiased estimators

Ex) $X_1, \dots, X_n$ be iid from $P(\lambda)$. Find MLE of λ

$$\left[\ f(x,\lambda) = \frac{e^{-\lambda}\lambda^x}{x!},\quad x \in 0, 1, \dots,\ \lambda > 0\ \right]$$

$$L(\lambda, \underline{x}) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{\Sigma x_i}}{\prod_{i=1}^{n} x_i!}$$

$$\text{Log}\left(L(\lambda, \underline{x})\right) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right) \ln \lambda - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

$$\frac{\partial \ln(\theta)}{\partial \lambda} = -n + \sum_{i=1}^{n} x_i / \lambda$$

$$\Rightarrow \hat{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\left.\frac{\partial^2 \ln(\lambda)}{\partial \lambda^2}\right|_{\hat{\lambda}} < 0 \quad \forall \ \lambda > 0$$

e.g.) Let $X_1, \dots, X_n$ be iid $N(\mu, \sigma^2)$ RVs

i) Find MLE of μ when σ known

ii) " " " σ^2 " μ "

iii) " " " μ & σ^2 (both unknown)

$\rightarrow$ same $\rightarrow L(\sigma^2, \underline{x})$

$L(\mu, \underline{x})$

$L(\mu, \sigma^2, \underline{x})$

Likelihood func^n of (μ, σ^2) given $\underline{x}$

$$L(\mu, \sigma^2, \underline{x}) = \prod_{i=1}^{n} f(x_i, \mu, \sigma^2)$$

$$= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$= \left(\frac{1}{\sqrt{2\pi}\,\sigma}\right)^n e^{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{n}(x_i-\mu)^2\right)}$$

1) log-likelihood of μ,

$$\log L(\mu, \underline{x}) = n \log\left(\frac{1}{\sqrt{2\pi}\,\sigma}\right) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

$$\frac{\partial}{\partial \mu} = 0 \Rightarrow \hat{\mu} = \bar{x}$$

2) $$\log L(\sigma^2, \underline{x}) = \ln\left(\frac{1}{\sqrt{2\pi}}\right)^n - \frac{n}{2}\ln \sigma^2 - \frac{1}{2\sigma^2}\left(\sum(x_i-\mu)^2\right)$$

$$\frac{\partial}{\partial \sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\Rightarrow \widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

E.g.– X_i be iid from $N(\mu, \sigma^2)$, both parameters unknown
find MLE of μ & σ^2

$$\underline{x} = (x_1, \dots, x_n) \qquad L(\mu, \sigma^2, \underline{x}) = \prod_{i=1}^{n} f_x(x_i, \mu, \sigma^2)$$

$$= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} = \frac{1}{(\sqrt{2\pi}\,\sigma)^n} e^{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{n} (x_i-\mu)^2\right)}$$

instead we maximize log-likelihood func :-

$$\ln(\mu, \sigma^2) = \log L(\mu, \sigma^2, \underline{x}) = n \ln\left(\frac{1}{\sqrt{2\pi}\,\sigma}\right) - \frac{n}{2} \ln\sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

$$\frac{\partial \ln(\mu, \sigma^2)}{\partial \mu} = 0 \qquad \frac{\partial(\ln(\mu, \sigma^2))}{\partial \sigma^2} = 0$$

$$\frac{2}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0 \quad \text{— (i)} \qquad \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (x_i - \mu)^2 = 0 \quad \text{— (ii)}$$

we need to solve (i) & (ii) simultaneously

$$\hat{\mu} = \bar{x} \qquad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2$$

put μ

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2 \neq \sigma^2$$

⊙ MLEs in general are not unbiased estimator

Invariance Property : Let $\hat{\theta}$ be MLE of θ. If $g(\theta)$ is some funⁿ of θ, MLE of $g(\theta) = g(\hat{\theta})$

$\downarrow$

not satisfied by unbiased estimators

e.g.- Let $\underline{X} \sim$ iid from $N(\mu, 1)$

$\hat{\mu} = \bar{X}$ $\quad (\hat{\mu^2}) = \bar{X}^2$ $\quad$ but $E(\mu^2) \neq \bar{X}^2$

e.g.- Let $\underline{X} \sim$ iid $N(\mu, \sigma^2)$, both μ and σ^2 are unknown.

MLE of $\sigma = ?$

$$\hat{\sigma} = \sqrt{\hat{\sigma}^2} = \sqrt{\frac{1}{n}\left(\Sigma (x_i - \bar{x})^2\right)}$$

$$\text{MLE of } \mu + \sigma = \widehat{\mu + \sigma} = \hat{\mu} + \hat{\sigma} = \bar{X} + \sqrt{\frac{1}{n}\Sigma(x_i - \bar{x})^2}$$

e.g.- Let $X_1, \ldots, X_n$ be iid from $G(\alpha, \beta)$. Find MLE of

i) only α unknown $\quad$ ii) only β unknow $\quad$ iii) both unknow

i) $$L(\alpha, \underline{x}) = \prod_{i=1}^{n} \frac{x_i^{\alpha-1} e^{-x_i/\beta}}{\beta^\alpha \Gamma\alpha}$$

$$= \frac{e^{-\Sigma x_i/\beta}}{\beta^{n\alpha} (\Gamma\alpha)^n} \prod_{i=1}^{n} x_i^{\alpha-1}$$

$$\ln(\alpha) = -n\alpha \ln\beta - n \ln \Gamma\alpha - \frac{\Sigma x_i}{\beta} + (\alpha-1)\Sigma \ln x_i$$

$$\frac{\partial}{\partial\alpha} = -n \ln\beta - \frac{n\Gamma'(\alpha)}{\Gamma\alpha} + \Sigma \ln x_i$$

$$\Rightarrow \quad \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = \frac{1}{n}\sum_{i=1}^{n} \ln x_i - \ln\beta$$

- We note that it is a non-linear eqn in α so we will be able to solve it only numerically

iii) $\ln(L(\alpha, \beta, \underline{x})) = -n\alpha \ln\beta - n \ln\Gamma\alpha - \frac{\sum x_i}{\beta} + (\alpha-1)\sum \ln x_i$

$$\frac{\partial}{\partial\beta} = \frac{-n\alpha}{\beta} + \frac{\sum x_i}{\beta^2} = 0 \quad \Rightarrow \quad \beta = \frac{\sum x_i}{n\alpha}$$

$$\frac{\partial}{\partial\alpha} = 0 \quad \Rightarrow \quad \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = \frac{1}{n}\sum_{i=1}^{n} \ln x_i - \ln\beta$$

$$\downarrow$$

$$\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = \left(\frac{1}{n}\sum \ln x_i - \ln\bar{x}\right) + \ln\alpha$$

$$\downarrow$$

$\hat{\beta} = \frac{\sum x_i}{n\hat{\alpha}}$ $\leftarrow$ need a numerical technique to solve, suppose we get $\hat{\alpha}$

—x—

Confidence Interval

—x—x—x—

- Till now we have discussed the method of point estimation be, unbiased, MLE.

- Sometimes, it is desired to have a range of values for the unknown parameter. We try to find an interval in which we can be highly confident that the unknown parameter lies in.

- Let $X_1, \dots, X_n$ be a random sample from a $f(x, \theta)$ distribution where $\theta \in \Theta$ is an unknown parameter

↓

Suppose $T_1(\underline{X})$ and $T_2(\underline{X})$ are any two statistics such that $P(T_1(\underline{X}) \le \theta \le T_2(\underline{X})) = 1-\alpha \quad \forall \theta \in \Theta$

$0 < \alpha < 1$

then the interval $[T_1(\underline{X}), T_2(\underline{X})]$ is called a $100(1-\alpha)\%$ confidence interval of θ.

↳ here $(1-\alpha)$ is the confidence coefficient

Result: Suppose $\underline{X}$ is a random sample from $\mathcal{N}(\mu, \sigma^2)$ distribution. Find $100(1-\alpha)\%$ confidence interval of

i) μ when σ known

ii) μ when σ unknown

- **Pivotal Quantity:** A func $T(\underline{X}, \theta)$ is said to be a pivotal Quantity for θ if prob. distrib^n of $T(\underline{X}, \theta)$ doesn't depend on θ.

i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$

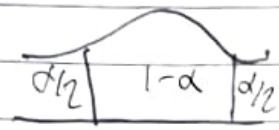
now consinder $\dfrac{\sqrt{n}(\bar{X}-\mu)}{\sigma} = Z$

$\Rightarrow Z \sim \mathcal{N}(0,1)$

Now, by area property of Z,

$P(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1-\alpha$

where $P(Z > z_{\alpha/2}) = \alpha/2$



$$Z = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$$

$$P\left(\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \le \mu \le \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) = 1-\alpha$$

$$\Rightarrow \left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2},\ \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right] = 100(1-\alpha)\%\ \text{CI of } \mu$$

↓

it is one of the interval, there can be many many others

This interval is a 2-sided symmetric $100(1-\alpha)\%$ CI of μ.

select z_a, z_b s.t. shaded $= 1-\alpha$

↓

we get a diff interval

$$\text{length of C.I.} = \frac{2\sigma}{\sqrt{n}} z_{\alpha/2}$$

ii) here $Z = \frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$ is not a pivotal quantity as Z is also a func of σ which is ~~another~~ unknown parameter other than μ.

suppose $T = \frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t(n-1)$

↓

pivotal quantity

Now by area property,

$$P(-t_{\alpha/2, n-1} \le T \le t_{\alpha/2, n-1}) = 1-\alpha$$

where $P(T > t_{\alpha/2, n-1}) = \alpha/2$

$$\Rightarrow P\left(\bar{X} - \frac{S}{\sqrt{n}} t_{\alpha/2, n-1} \le \mu \le \bar{X} + \frac{S}{\sqrt{n}} t_{\alpha/2, n-1}\right) = 1-\alpha \quad \forall \mu \in \mathbb{R}$$

↓

$\left[\bar{X} - \frac{S}{\sqrt{n}} t_{\alpha/2, n-1},\ \bar{X} + \frac{S}{\sqrt{n}} t_{\alpha/2, n-1}\right]$ is $100(1-\alpha)\%$ C.I.

Interval length $= \frac{2S}{\sqrt{n}} t_{\alpha/2, n-1}$ → (random) as S is there

• Most often, $\alpha \to 0.05, 0.01, 0.1$

E.g.:— Suppose length of life of a electric bulb $\sim N(\mu, 1296)$. If a random sample of size $n = 27$ bulbs are tested with $\bar{x} = 1478$ hrs. Find 95% interval of μ

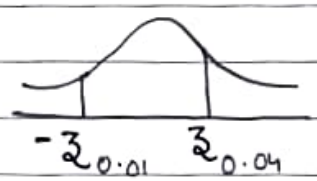
$$z_{0.025} = 1.96, \quad \sigma = 1296, \quad n = 27$$

~~from~~ a 2-sided $100(1-\alpha)\%$ symmetric CI of μ is $\left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{0.025}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{0.025}\right]$

$[1464.42, 1491.58]$ is a 95% CI of μ

↳ length $= 27.16$

For fun, let's take another 95% C.I. of μ



$\to \left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{0.01}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{0.04}\right]$

↓

$[1461.85, 1490.19]$

↓

length $= 28.34$

(more)

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E.g.– Suppose the amount of butterfat in pounds – μ & σ^2 unknown. $n = 20$, $\underline{X}$:–

481, 537, 513, 583, 453, 510, 570, 500, 457, 555, 618, 327, 350, 643, 499, 421, 505, 637, 599, 392.

Find a 95% CI of μ.

$\hat{X} = \frac{\Sigma x_i}{n=20}$ $\qquad$ $S = \sqrt{\frac{1}{19}(\Sigma(x_i - \bar{x})^2}$

$\bar{X} = 507.5$, $\qquad$ $S = 89.57$

E.g.– The average zinc conc. recorded in 36 different locations = 2.6 g/ml. Find 95% and 99% CI for mean zin conc. in river. ($\sigma = 0.3$ given)

95% $\rightarrow$ [2.50, 2.70]
99% $\rightarrow$ [2.47, 2.73]

• Suppose if we estimate μ by $\bar{x}$ then error e is defined

$$P(|\bar{x} - \mu| \leq e) = 1 - \alpha$$

$$\Rightarrow P\left(|z| \leq \frac{\sqrt{n}\, e}{\sigma}\right) = 1 - \alpha$$

$$\Rightarrow \frac{\sqrt{n}\, e}{\sigma} = z_{\alpha/2}$$

$$\Rightarrow n = \left(\frac{z_{\alpha/2}\, \sigma}{e}\right)^2$$

e.g- In previous example how large a sample is req. if we want to 95% confident that μ is off by less than 0.05

$$P(|x-\mu| \leq 0.05) = 0.95$$

$$z_{0.025} = 1.96, \quad e = 0.05, \quad \sigma = 0.3$$

$$n = \left(\frac{0.3 \times 1.96}{0.05}\right)^2 = 138.3 \approx 139$$

—x—

Hypothesis Testing

- Let $X_1, \ldots, X_n$ be iid from $f(x, \theta)$ distrib^n where $\theta \in \Theta$ is an unknown parameter.

Like estimation methods, hypothesis testing is also useful making some interference abt θ based on $\underline{X}$

↓

- A statistical hypothesis is some statement about unknown parame-ter θ.

e.x- $X \sim \mathcal{N}(\theta, 1)$

$\theta = 1$ $\quad$ $\theta > 100$

$\theta = 10$ $\quad$ $\theta \leq 100$

you can make any statement

⊙ **Simple Hypothesis :** A statistical hypothesis that completely specify a prob distribⁿ called simple hypothesis.

⊙ **Composite Hypothesis :** A statistical hypothesis which does not completely specify a prob distribⁿ under consideration.

$X \sim N(\theta, 1)$	$X \sim N(\theta, \sigma^2)$	$X \sim N(\theta, 1)$
Simple $\rightarrow \theta = 10$	composite $\rightarrow \theta = 10$	composite $\rightarrow \theta \leq 1$
	Simple $\rightarrow \begin{cases} \theta = 10 \\ \sigma = 1 \end{cases}$	

⊙ **Test of Hypothesis :** A procedure that satisfies a set of rules for decision whether to accept / reject the hypothesis under consideration.

→ **Null Hypothesis :** A hypothesis which is setup (or assumed) and whose value is tested for possible rejection on the basis of $X_1, \dots, X_n$ drawn from $f(x, \theta)$ distribⁿ, often denoted as H_0.

→ **Alternative Hypothesis :** Hypothesis that differ from null hypothesis. Often denoted as H_i. The alternative hypothesis is not tested but its acceptance / rejection depends on acceptance / rejection of H_0.

↓

presents a contradictory statement to null hypothesis.

- **Test statistic**: A $f(\underline{X})$ whose computed value determines the final decision of acceptance/rejection of H_0.

↳ **Critical Region**: The set of values of the test statistic which leads to rejection of H_0 is called CR of the test.

Level of Significance: The max^m prob^{ty} with which a true null hypothesis is rejected is known as level of significance.
Denoted as α ($\alpha \to 0.05, 0.01, 0.1$)

- How to state a Hypo^{ths} test problem?
  ↳ $H_0 : \theta \in \Theta_0 \subset \Theta$ where $\Theta_0 \cap \Theta_1 = \phi$
  $H_1 : \theta \in \Theta_1 \subset \Theta$ $\Theta_0 \cup \Theta_1 \subseteq \Theta$

then based on evidence collect^{ed} from a random sample $X_1, \ldots, X_n$ from a $f(x, \theta)$ distrib^n, the problem is to select one of the two hypothesis.

- Since we make decision to accept/reject H_0 on the basis of random sample so it is possible that some errors may occur.

Decision based on $\underline{X}$	True Situation	
	H_0	H_1
Accept H_0 / Do not reject H_0	correct	error Type II error
Reject H_0	error Type I error	correct

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Type 1 error
↓
when we reject H_0 by a test even if it is true.
↓
often CR is determined st prob. of type 1 error is equal to level of significance
↓
$\alpha = P(\text{type 1 error})$
$= P(\text{reject } H_0 / H_0 \text{ true})$

Type 2 error
↓
When we accept H_0 by a test even if it is false
↓
$\beta = P\left(\text{Accept } H_0 \text{ or dont reject } H_0 / H_0 \text{ false or } H_1 \text{ true}\right)$

Eg — Let $X_1, \dots, X_9$ be iid $\mathcal{N}(\theta, 1)$ RVs. Test the hypothesis $H_0 : \theta = 5.5$ v/s $H_1 : \theta = 8$

<u>Test 1:</u> Reject H_0: iff $X_1 > 7$

<u>Test 2:</u> Reject H_0 iif $\frac{X_1 + X_2}{2} > 7$

<u>Test 3:</u> Reject H_0 iif $\bar{X} > 6$

<u>Test 4:</u> Reject H_0 iif $\bar{X} > 7.5$

For test 1, $\alpha = P(X_1 > 7 / \theta = 5.5) = P(Z > 1.5)$
$= 0.06681$

$\beta = P(X_1 \leq 7 / \theta = 8) = P(Z \leq -1)$
$= 0.15866$

Test	1	2	3	4
α	0.06681	0.01696	0.06681	0
β	0.15866	0.07865	0	0.06681

2 better as both errors are less

can't say

In general if we try to dec one error other will tend to inc & vice versa.

- In practice one error is fixed and then the other one is minimized. Commonly α is specific.

Testing mean of normal distribn

- Let $X_1, \dots, X_n$ be iid from $\mathcal{N}(\theta, \sigma^2)$, θ: unknown, σ: known

Suppose we are interested in testing the null hypothesis

(1) $H_0 : \mu = \mu_0$ and $H_1 : \mu > \mu_0$

where μ_0 is a give constant

when level of significance is α

$\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$ is a good estimator for μ

$\downarrow$

It may seem reasonable to accept H_0 if value of $\bar{X}$ is not too far from μ_0

$\downarrow$

Accordingly the CR is defined as

$CR : \{(x_1, \dots, x_n) : \bar{x} - \mu_0 > c\}$ for some suitable chosen c

If we desire our test to have significance α we must find c s.t.

$$P(\bar{x} - \mu_0 > c \,/\, H_0 \text{ true}) = \alpha$$

$$P_{\mu_0}(\bar{X} - \mu_0 > c) = \alpha$$

under H_0, $\bar{X} \sim N(\mu_0, \sigma^2/n)$

$$P\left(Z > \frac{c\sqrt{n}}{\sigma}\right) = \alpha$$

$$\Rightarrow \frac{c\sqrt{n}}{\sigma} = z_\alpha$$

$$\Rightarrow c = \frac{z_\alpha}{\sqrt{n}}\sigma$$

Thus level of significance α is to reject H_0 if $\bar{X} - \mu_0 > \frac{z_\alpha}{\sqrt{n}}\sigma$

or if $Z > z_\alpha$

(II) $H_0 : \mu = \mu_0$

$H_1 : \mu < \mu_0$

$\downarrow$

$Z < -z_\alpha$ or $\bar{X} < \mu_0 - \frac{\sigma}{\sqrt{n}} z_\alpha$

(III) $H_0 : \mu = \mu_0$

$H_1 : \mu \neq \mu_0$

$\hookrightarrow |Z| > z_{\alpha/2} \rightarrow \bar{X} > \mu_0 + \frac{z_{\alpha/2}}{\sqrt{n}}\sigma$

$\bar{X} < \mu_0 - \frac{z_{\alpha/2}}{\sqrt{n}}\sigma$

E.g.— In a college, there are many sections of a particular subject. In common exam the avg mark follow $N(66, 24 \times 24)$ distribn. In a section of 32 stodents, the avg marks turned out as 53. We want to test the hypothesis whether the given section comes from the same population or with a lower avg class. Take $\alpha = 0.05$

$H_0 : \mu = 66$
$H_1 : \mu < 66$

Test statistics $Z = \dfrac{\sqrt{n}\,(\bar{x} - \mu)}{\sigma}$

under H_0, $Z = \dfrac{\sqrt{n}\,(\bar{x} - \mu_0)}{\sigma} \sim N(0,1)$

computed value of $Z = \dfrac{\sqrt{32}\,(53 - 66)}{24}$

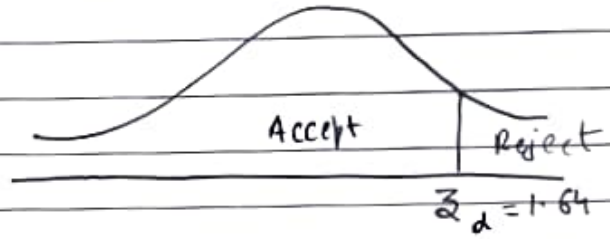
$= -3.055$

A 5% level of significance from Z table $= -1.65$

$\Rightarrow$ Reject H_0

E.g.– In a large farm, the cows weigh on avg 520 kg. A special diet is tried on 50 randomly selected cow and their weight is observed to have an avg of 528 kg. Assume that popul^n is $\mathcal{N}$ with S.D. = 25kg. Test the hypothesis if cow weight can ↑ (with $\alpha = 0.05$)

$H_0 : \mu = 520$
$H_1 : \mu > 520$



TS = 2.26

Reject H_0 at 5% level of significance

E.g.– The weight of pots made by a standard process ~ $\mathcal{N}(345, (2.8)^2)$. A pot produced just before the process closed before the day weight 338.5. Use 5% level of sig. to test the hypothesis if process worked correctly

$H_0 : \mu = 345$
$H_1 : \mu \neq 345$ → if no info. given take two sided

Reject H_0 if $|Z| > Z_{0.025}$

$$\begin{cases} Z > 1.96 \\ Z < -1.96 \end{cases}$$